Experiment No. 3

Name of experiment:

1. ***Relationship between speed and induced voltage for <u>separately excited</u> D.C generator.***
2. ***Relationship between speed and induced voltage for <u>self-excited</u> D.C generator***

1. Object:

This test is done to obtain the curve between speed and voltage speed\voltage characteristic of D.C separately excited and self-exited D.C generator.

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2. Theory:

The induce voltage in D.C generator is given by:

$$Eg = \frac{Z\Phi PN}{60\,A}$$

As Z, P&A are constants, the equation becomes ((Eg =CΦN)) I_f Φ is constant, then

((E α N)) ---------- (1)

From equation (1) if Φ is kept constant (i.e the field current is constant), we can obtain the relationship between the speed and the voltage. This characteristic is a linear relation for separately excited D.C generator and nonlinear for self-exited dc generator.

3. Procedure:

1. Connect the circuit as shown in figure (1),(2).
2. Adjust the speed at 30% of the rated value, Keeping the field current constant.
3. Increase the speed gradually in different steps till it reaches 120% of the rated value.
4. Record the voltage induced across the armature terminals (be sure that at each Reading the field current is constant).

1. ***Relationship between speed and induced voltage for <u>separately excited</u> D.C generator.***

I_f =0.5A							
N(r.p.m)	200	400	600	800	1000	1200	1400
E_g(volt)							

I_f =0.25A							
N(r.p.m)	200	400	600	800	1000	1200	1400
E_g(volt)							

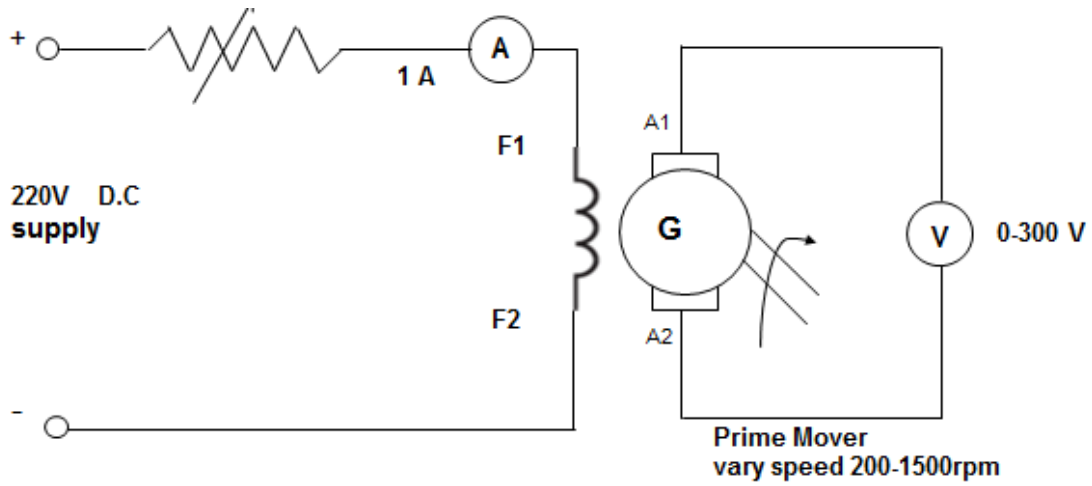


Fig (1)

2. ***Relationship between speed and induced voltage for <u>self-excited</u> D.C generator***

<u>The field current constant with the variation of speed</u>

I_f =0.5A							
N(r.p.m)	200	400	600	800	1000	1200	1400
E_g(volt)							

<u>The field current vary with the variation of speed</u>

N(r.p.m)	200	400	600	800	1000	1200	1400
I_f =0.5A							
E_g(volt)							

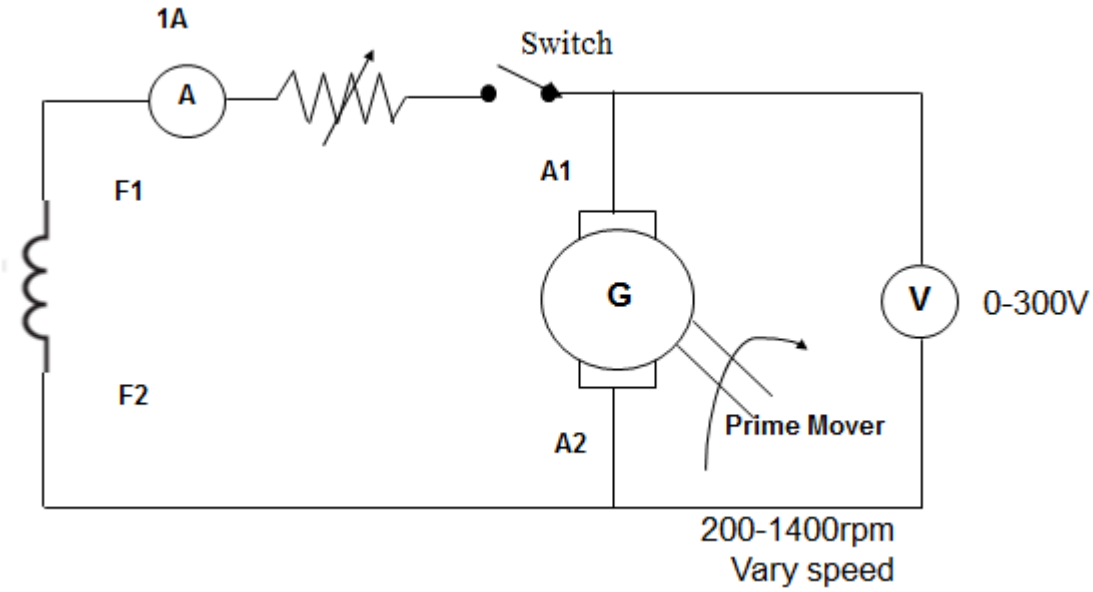


Fig (2)

5. Report:

1. Plot the speed /Voltage curve. For both type and both case.
2. What are the conditions for generating E.M.F voltage (self & separately generator)?
3. Is the induced voltage in conductors of armature A.C or D.C? What is its frequency and what is the relationship between speed and frequency?
4. What is the function of armature core?
5. The E emf generated by shunt DC generator is E. now while pole & flux remains constant. If the speed of the generator is *doubled*, how will be the emf generated
6. Discuss your results clearly and briefly.
7. How we can reversing the polarity of DC generator

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